

Best Directions

Energy gains and losses through windows are a strong function of a particular window's U-factor and solar heat gain coefficient (SHGC), of course, but orientation plays a critical role as well. Far too often, for my taste, I've observed that the facade with the greatest number of windows on a pro-

factors and SHGCs make them much better candidates for both new and retrofit applications. Single-glazed clear windows are no longer used in new construction, but unhappily, they are still found on many older homes in the Southwest, as are double-glazed clear windows with high SHGCs.

with an overall coefficient of performance (COP) of 3, 40 kWh of cooling would be necessary to meet the window load on this single bright day in July.

Double-glazed windows with a low SHGC (0.38) and low U-factor (0.3) are small net losers in all directions on the bright January day, because no

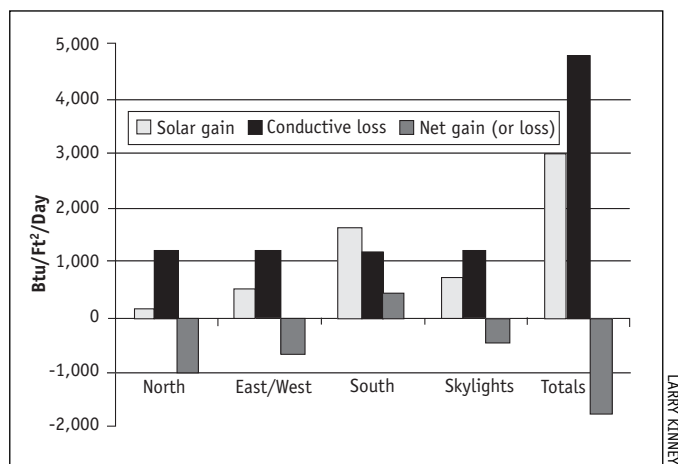


Figure 1. January 21, single glazing, SHGC = 0.9; U = 1.0; loss = 1,830 Btu/ft²/day.

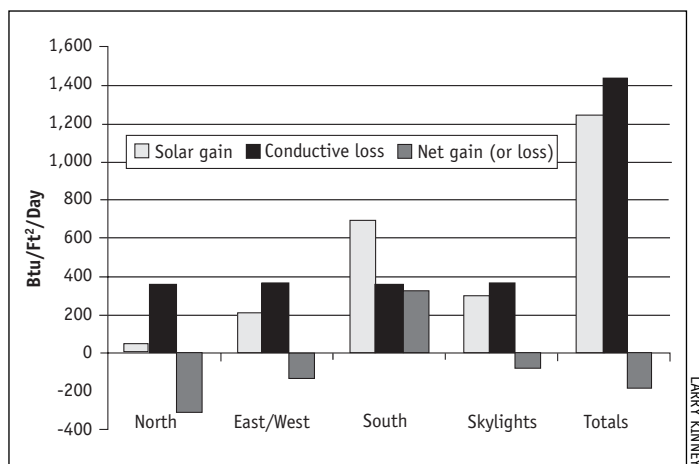


Figure 2. January 21, double glazing, gas fill, SHGC = 0.38; U = 0.3; loss = 186 Btu/ft²/day.

duction built home tends to face the backyard, whatever its orientation.

To get a feel for just how much of an effect a window's orientation can have, I modeled the energy impacts of placing three different types of windows in four separate locations on both a summer and a winter day. All six of the bar charts shown in Figures 1–6 depict solar gains and conductive losses in Btu/ft² for a single 24-hour day with clear skies at 40° north latitude (approximately the latitude of Denver, Reno, and Salt Lake City). Figures 1–3 depict conditions for January 21, with an average temperature of 20°F. Figures 4–6 depict conditions for July 21, with an average temperature of 85°F. Net gains help in the winter and hurt in the summer.

The data for the single-glazed clear window are shown here to illustrate the difference between its performance and the performance of windows whose U-

These plots reveal a number of trends. (Note that the y-axes in the plots for Figures 1–6 have different calibrations.)

Single-glazed windows with a high SHGC (0.9) and U-factor (1.0) are net losers in all directions both summer and winter, except on the south facade in the winter (Figures 1 and 4).

They are particularly poor performers in the summer. With 270 ft² of evenly distributed glazing and a cooling system

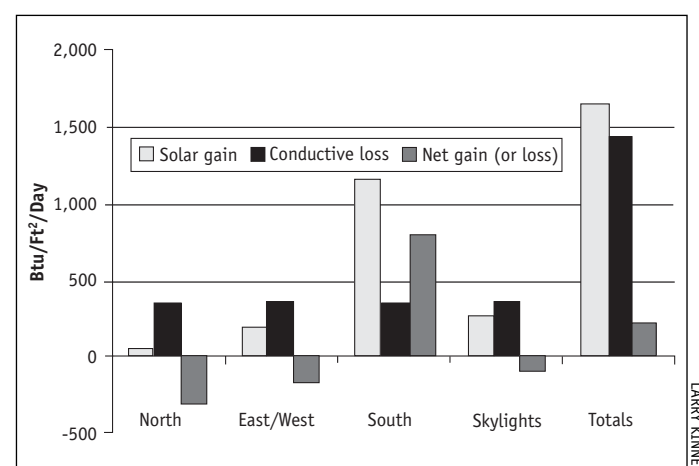


Figure 3. January 21, double glazing, gas fill, SHGC = 0.38 (South 0.72); U = 0.3, gain = 220 Btu/ft²/day.

facade except for the south has enough solar gain to make up for losses (Figures 2 and 5). However, net losses in the

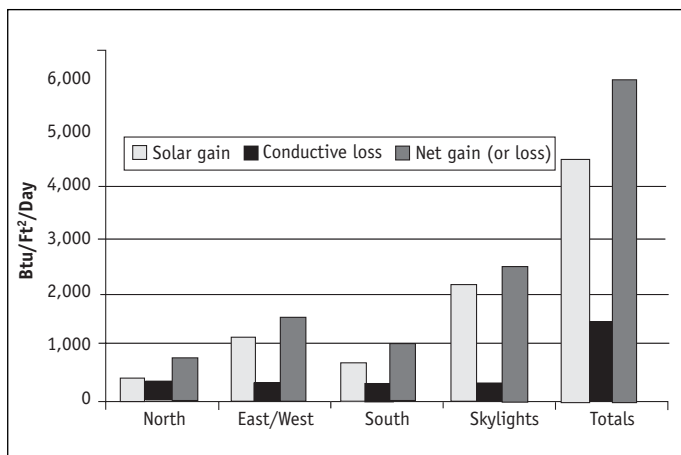


Figure 4. July 21, single glazing, SHGC = 0.9; U = 1.0; gain = 5,931 Btu/ft²/day.

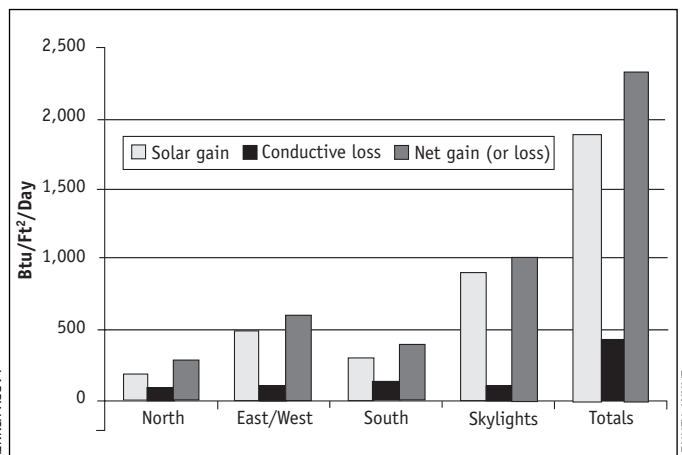


Figure 5. July 21, double glazing, gas fill, SHGC = 0.38; U = 0.3; gain = 2,328 Btu/ft²/day.

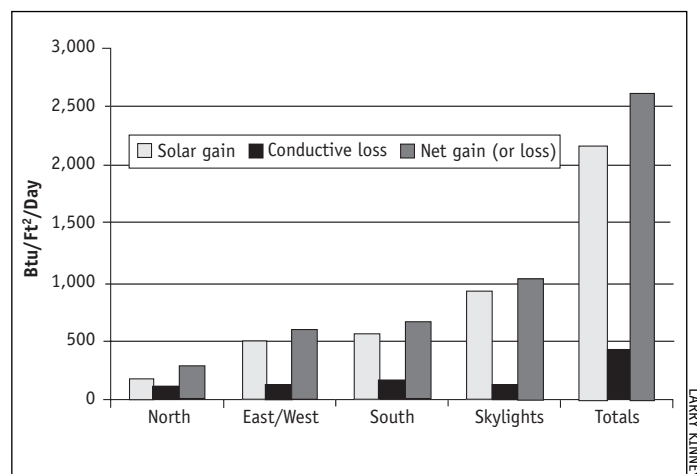


Figure 6. July 21, double glazing, gas fill, SHGC = 0.38 (South 0.72), U = 0.3; gain = 2,594 Btu/ft²/day.

Table 1. The Effect on July 1 of Adding 90% Effective Shading Devices to the Window Systems in Figures 4, 5 and 6.

Parameter	U = 0.1 SHGC = 0.9	U = 0.3 SHGC = 0.38	U = 0.3 SHGC = 0.38 (South SHGC = 0.72)
Solar Gain (Btu/ft²/day)	2,300	990	1,300
Conductive Gain (Btu/ft²/day)	1,100	320	320
Total Gain (Btu/ft²/day)	3,400	1,300	1,600
Total Gain w/ Shading (Btu/ft²/day)	1,300	420	450
Savings (Btu/ft²/day)	2,100	880	1,200
Savings (%)	62%	68%	75%

winter are a factor of 10 less than net losses in the winter for single-glazed windows. The summertime cooling load is about 40% that of the single-glazed case; 16 kWh of cooling energy would be required for the bright day in July.

The case shown in Figures 3 and 6 has the same glazing as the case shown in Figures 2 and 3, but the south-facing glazing has a SHGC of 0.72 instead of 0.38. This gives better wintertime performance, allowing the entire glazing system to produce a net gain. There is a slight penalty paid in the summer, of course; 17 kWh of cooling energy would be required for the bright day in July—1 kWh more than in the case with low SHGC glazing.

Skylights are net thermal losers in the winter and account for quite substantial solar gains in the summer. Exterior netting in the summer can limit solar gain while retaining a measure of natural illumination.

It is useful to examine the effect of adding shading devices for the midsummer case of the three windows systems. Table 1 assumes that such devices are 90% effective in shading direct-beam sunlight. This analysis shows that adding overhangs, awnings, fins, or other external shading devices (including trees and vines) to block direct-beam sunshine substantially lowers the cooling loads for all glazing types. Of course, the absolute

beneficial effect of shading devices is most pronounced for window systems with particularly high SHGCs—approximately twice that of the more efficient glazing systems.

These considerations show that judicious use of overhangs and other external shading devices, in combination with window SHGCs tuned to direction, can produce excellent overall performance.



—Larry Kinney

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